Introduction to Machine Learning

Lecture 3 — Supervised Learning

Dr. Priya Nair · CS 4810 · Spring 2026

Today's agenda

1. Recap
2. What is a model?
3. Types of learning
4. Supervised Learning
5. The learning loop
6. Linear regression — intuition
7. Linear regression — in Python
8. Decision trees — intuition
9. Comparing models
10. Overfitting
11. What is overfitting?
12. Bias–variance tradeoff
13. Remedies for overfitting
14. Quick check
15. Lab preview
16. Lab environment walkthrough
17. Resources

Recap

What is a model?

> *A model is a function that maps inputs to outputs — learned from data rather than hand-coded.*
> *— Bishop, Pattern Recognition and Machine Learning*

Types of learning

Supervised — labelled training data

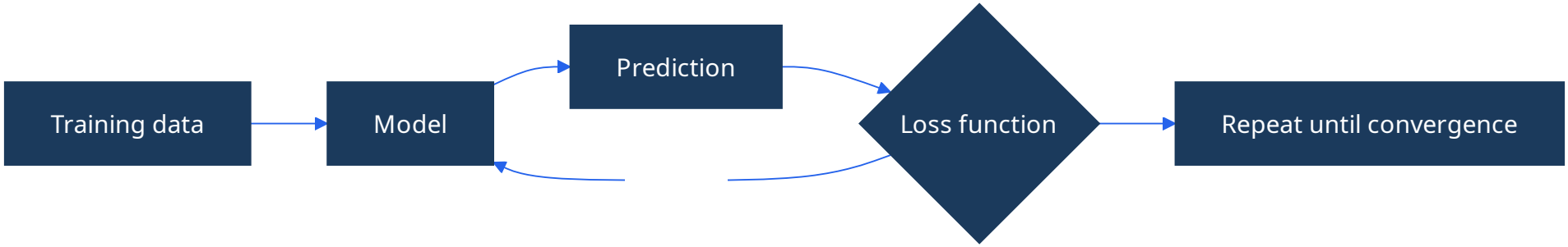
- Classification: "Is this email spam?"
- Regression: "What will the price be?"

Unsupervised — no labels

- Clustering: "Which customers are similar?"
- Dimensionality reduction: "What's the structure?"

Supervised Learning

The learning loop



Linear regression — intuition

We want to find the line of best fit through our data points.

Formally, we learn parameters **w** and **b** such that:

$$\hat{y} = w \cdot x + b$$

minimises the **mean squared error** over all training examples:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Linear regression — in Python

PYTHON

```python
from sklearn.linear_model import LinearRegression
import numpy as np

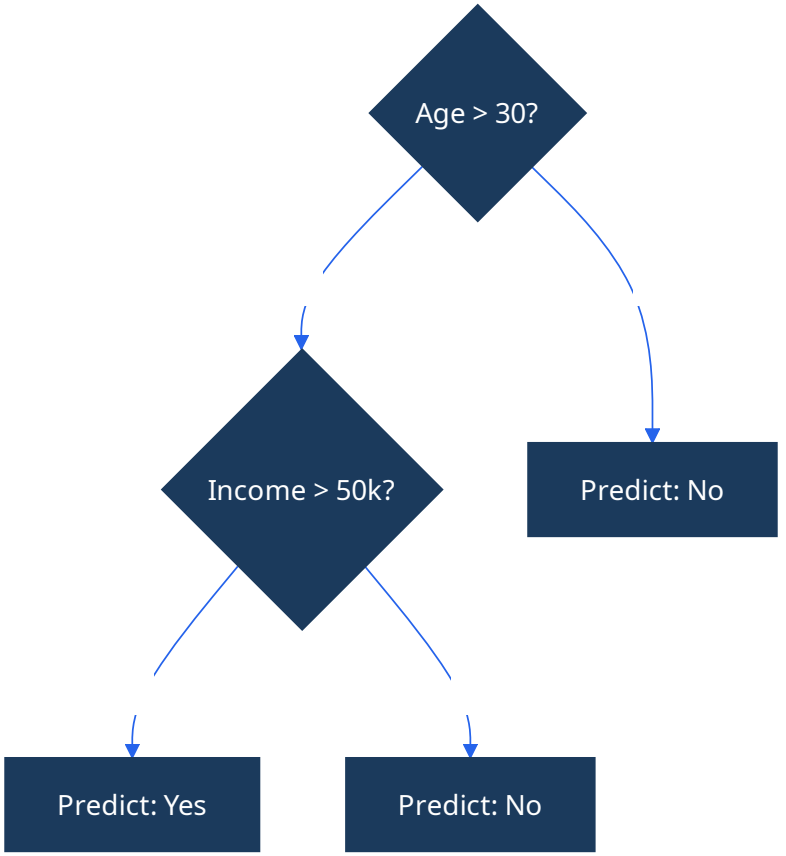
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2.1, 3.9, 6.2, 7.8, 10.1])

model = LinearRegression()
model.fit(X, y)

print(f"Slope:     {model.coef_[0]:.2f}")
print(f"Intercept: {model.intercept_:.2f}")
print(f"Predicted for x=6: {model.predict([[6]])[0]:.2f}")
```

Decision trees — intuition

A decision tree partitions the feature space using a series of binary questions. Each internal node is a feature split; each leaf is a prediction.



Comparing models

	Linear regression	Decision tree
Output type	Continuous	Continuous or categorical
Interpretable	✅ Easy	✅ Easy
Non-linear relationships	❌	✅
Sensitive to outliers	✅ Yes	Less so
Prone to overfitting	Rarely	Very much so

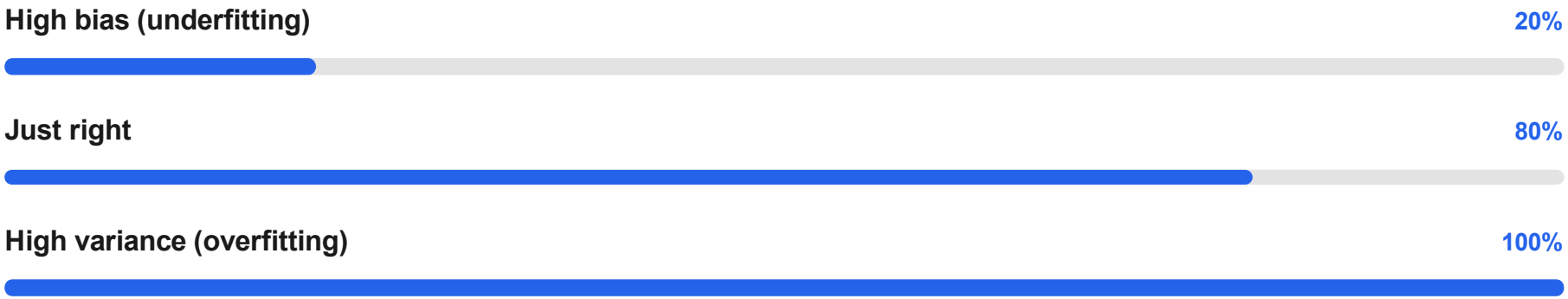
Overfitting

What is overfitting?

> *A model overfits when it learns the training data so well that it fails to generalise to new examples.*

The model has memorised noise, not signal.

Bias–variance tradeoff



Remedies for overfitting

Decision trees

- Limit max depth (`max_depth=4`)
- Require minimum samples per leaf
- Prune after training

General techniques

- More training data
- Cross-validation
- Regularisation (L1 / L2)
- Ensemble methods (next week)

Quick check

Which model would you use for predicting house prices?

https://pollev.com/priya-nair-cs4810

Click to open in browser

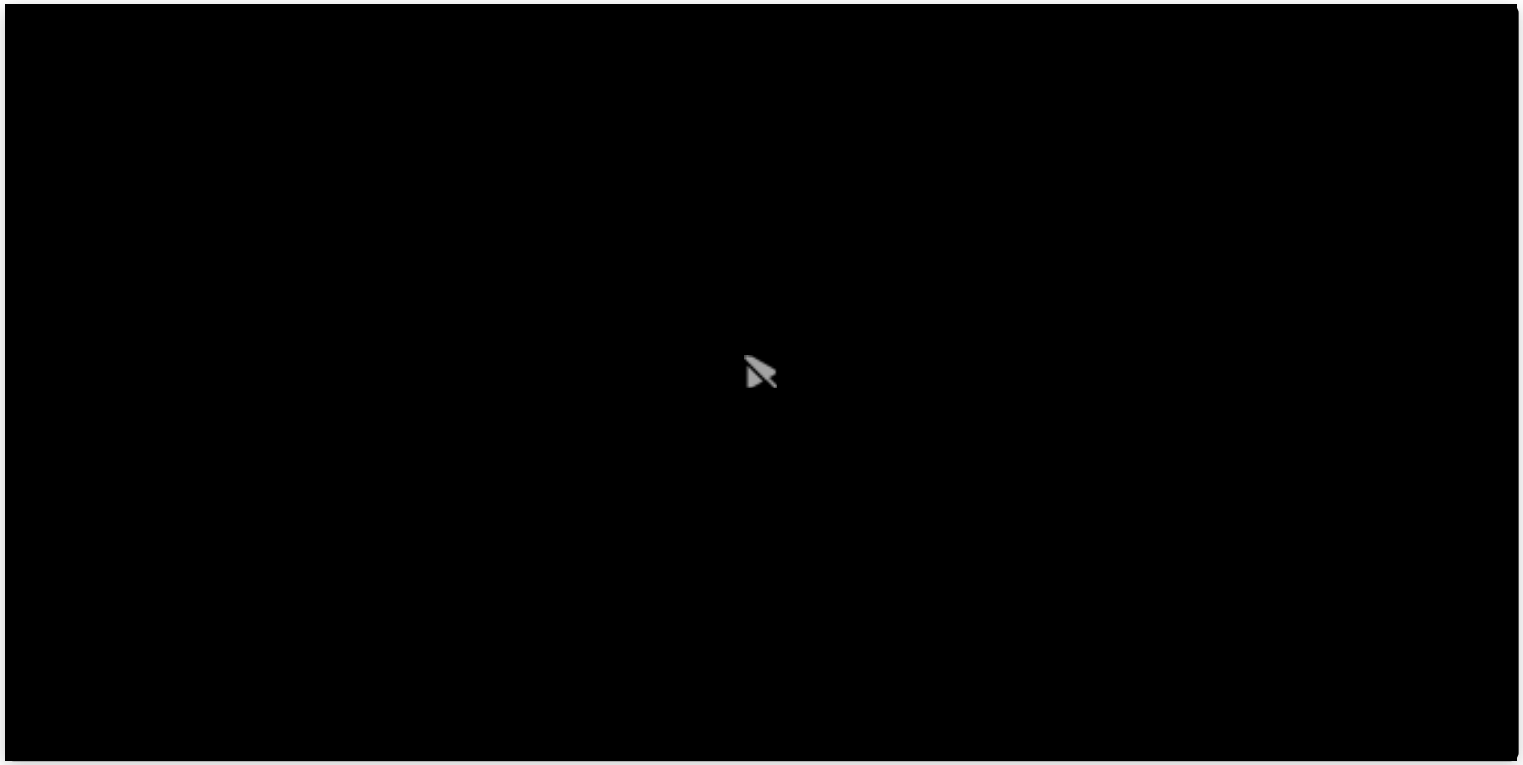
Lab preview

In today's lab you will:

- Load the Boston Housing dataset
- Fit a linear regression model and a decision tree
- Evaluate both on a held-out test set
- Plot the bias–variance curves

> ⚠ **Deadline**
>
> Submit your notebook to Gradescope by **Friday 23:59**.

Lab environment walkthrough



Lab environment walkthrough

Resources

- Course notes: Chapter 3 — Linear Models
- Scikit-learn docs: `sklearn.linear_model`, `sklearn.tree`

See you Thursday.

Bishop, C. (2006). Pattern Recognition and Machine Learning. Springer, Ch. 3.1–3.3.